

# Session 20: AI Bias & Fairness

You've learned how AI reads, sees, hears, and combines senses. You've explored how AI is reshaping industries and creating new careers. Now it's time to ask a critical question: **Is AI fair?** Today, you'll discover where bias comes from, how it causes real harm, and what we can do to build fairer AI for everyone.



# Objectives

01

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## Define AI Bias

Understand what AI bias is and why it matters for everyone.

02

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## Identify Sources

Recognize the three main sources of bias in AI systems.

03

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## Real-World Consequences

Recognize how biased AI causes documented harm to real people.

04

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## Detect & Reduce Bias

Explore strategies for finding and mitigating bias in AI.

05

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## Human-in-the-Loop

Apply the Human-in-the-Loop approach to fairness in AI decisions.

# Recap: AI Careers & The Future of Work

| Concept                | Key Takeaway                                                                            |
|------------------------|-----------------------------------------------------------------------------------------|
| AI & Work              | AI automates repetitive tasks, augments human skills, and creates entirely new careers. |
| Essential Skills       | Thinking Skills plus Human Skills equal your AI-era superpowers.                        |
| Four Career Lanes      | Builders, Deciders, Guardians, and Empowered.                                           |
| Human-AI Collaboration | AI handles scale and speed; humans provide wisdom and oversight.                        |



# Warm-Up: Who Gets Approved?

Two people apply for a small business loan at a local bank. An AI system makes the decision.

## Casey

- Lives in a neighborhood with high property values
- Graduated from a private university
- Loan application: **Approved** ✓

## Riley

- Lives in a neighborhood with lower property values
- Has the **same job, same income, and same credit score**
- Loan application: **Denied** ✗

Does this outcome feel fair? Why or why not?

What might the AI have learned from its training data?

Who should be responsible for this decision?



## What Is AI Bias?

AI bias happens when an artificial intelligence system produces results that are **systematically unfair** to certain groups of people. The word "systematically" is important - it means the unfairness isn't a random glitch or a one-time mistake. It's built into the way the AI operates, and it keeps happening over and over again.



# What Is AI Bias?

## Not Intentional

AI has no feelings or intentions.  
Bias comes from patterns in data  
and design choices made by  
humans.

## Not Aware

AI doesn't "know" it's being unfair.  
It's just following mathematical  
patterns it learned.

## A Reflection

AI is a pattern-matching engine. It  
reflects the world it was trained on  
- including the world's unfairness.

 AI is a pattern-matching engine, not a thinking being. It reflects the world it was trained on - including the world's unfairness.

# Where Does Bias Come From?

Bias in AI doesn't appear out of nowhere. It flows from **three main sources**. Understanding each source helps us know where to look for problems.

## Source 1: Data Bias

The training data doesn't represent everyone equally. Some groups are overrepresented; others are barely present or completely missing.

## Source 2: Algorithm Bias

Even with good data, the way the AI is designed to learn can introduce or amplify bias by latching onto unfair patterns.

## Source 3: Human Bias

The people who build AI systems have unconscious biases that seep into the AI through data collection, labeling, and design choices.

# Source 1: Data Bias

The training data used to teach the AI doesn't represent everyone equally. Some groups are overrepresented; others are barely present or completely missing. Since AI learns only from what it's shown, it becomes an expert on the groups it sees most often and fails on the groups it rarely sees.

## Example

Imagine learning to cook by watching only Italian cooking shows. You'd become great at pasta and pizza but if someone asked you to prepare a dish from Thailand or Mexico, you'd be completely lost. You weren't shown those examples. **Data bias works exactly the same way.**

## Example

A photo dataset labeled "outdoor activities" contains mostly images of people from one region. The AI learns that "outdoor activity" looks like that specific group in that specific landscape. When shown a photo of a family having a picnic in a different setting, the AI may fail to recognize it.



## Source 2: Algorithm Bias

Even with good data, the way the AI is designed to learn, the algorithm itself can introduce or amplify bias. Some algorithms are designed to find the simplest, strongest pattern. If that pattern happens to be correlated with something unfair, the algorithm will latch onto it because it's **mathematically efficient, not because it's fair**.

### Example

Imagine a talent scout who only watches games played in large stadiums. They might conclude that the best athletes all come from big cities, completely missing incredible talent from smaller towns. **The scout's method created the bias.**

### Example

A hiring algorithm trained on past successful employees may learn that candidates who use certain phrases get higher scores. If those phrases were more common in resumes from one gender or background, the algorithm will favor that group - even if the phrases have nothing to do with job performance.

## Source 3: Human Bias

The people who build AI systems - who collect the data, label the examples, and decide what the AI should optimize for - are human. Humans have unconscious biases. Those biases can seep into the AI through the choices we make about what data to collect, how to label it, and what problem we ask the AI to solve.

### Example

Imagine a teacher writing a test paper and unintentionally using examples that only students who play a certain sport would understand. The test isn't designed to be unfair, but students who don't play that sport are at a disadvantage. **Human bias in AI is similar - it's often unintentional but still causes harm.**

### Example

Developers building an AI decide to include "distance from a major city" as a factor, thinking it might relate to job opportunities. But this choice unintentionally disadvantages people in rural areas, even if their financial situation is identical to someone in a city.

# Mini Activity 1: Spot the Source of Bias

Read each short scenario carefully. For each one, decide whether the primary source of bias is **Data Bias**, **Algorithm Bias**, or **Human Bias**. Be prepared to explain your reasoning.

| # | Scenario                                                                                                                                                                                     | Your Answer |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 1 | A voice assistant struggles to understand speakers with regional dialects because its training recordings came almost entirely from one metropolitan area.                                   |             |
| 2 | A resume-filtering AI was trained on ten years of successful employees who were mostly from a single demographic group. It now assigns higher scores to applicants with similar backgrounds. |             |
| 3 | A photo-tagging system labels anyone standing near a stroller as "parent" and often misidentifies caregivers who are not women.                                                              |             |
| 4 | Developers include "proximity to a major transit station" as a factor in a loan approval model, unintentionally disadvantaging applicants in rural communities.                              |             |
| 5 | A translation tool always renders the word for "nurse" as female and "pilot" as male in languages with gendered nouns because its training books reflected those stereotypes.                |             |

# Mini Activity 1: Answer Key

| # | Scenario                              | Primary Source | Why?                                                                                                   |
|---|---------------------------------------|----------------|--------------------------------------------------------------------------------------------------------|
| 1 | Voice assistant and regional dialects | Data Bias      | The training data lacked diverse voices. The AI never heard those dialects, so it couldn't learn them. |
| 2 | Resume AI favors one demographic      | Data Bias      | The training data reflected past hiring patterns that were not diverse. The AI learned from history.   |
| 3 | Photo-tagging near strollers          | Data Bias      | Training images likely contained stereotypical associations between women and caregiving.              |
| 4 | Transit factor in loan model          | Human Bias     | Developers made a design choice that seemed logical but had unintended unfair consequences.            |
| 5 | Gendered translation of professions   | Data Bias      | Training books contained historical gender imbalances in how professions were portrayed.               |



# Bias Detection: Finding Hidden Unfairness

You can't fix a problem you can't see. Before we can make AI fairer, we have to find the bias. This process is called **bias detection**. It involves asking three essential questions and using specific techniques to get answers.

## Who is in the training data?

And who is left out?

## Does the AI perform equally well?

For different groups of people?

## Are the labels and features fair?

Are they appropriate for all groups?

# Bias Detection Techniques



## Data Audit

Carefully examine the training dataset to see who and what is represented. Are certain groups overrepresented? Are others missing entirely?



## Performance Testing

Run the AI on different groups of people separately and measure how accurate it is for each group. Does the system work equally well for everyone?



## Disaggregated Metrics

Break down overall performance numbers by group. Instead of "95% accurate overall," ask: "What is the accuracy for Group A? For Group B?"

**i** You cannot fix what you do not measure. Bias detection is the essential first step toward building fairer AI.



# Bias Mitigation: Making AI Fairer

Once we've detected bias, the next step is **mitigation** - taking action to reduce or eliminate the unfairness. There is no single magic button. Mitigation requires a combination of strategies applied at different stages of the AI lifecycle.

## Diverse Data

Collect training data that genuinely reflects the diversity of people who will use the AI. Intentionally include underrepresented groups. *Apply before training.*

## Ongoing Testing

Bias testing shouldn't happen only once. Regularly test the AI on different demographic groups. If a new version is released, test it again. *Apply continuously.*

## Human-in-the-Loop

For high-stakes decisions - hiring, loans, healthcare - a human must review AI recommendations before a final decision is made. *Apply before life-affecting decisions.*

## Transparency

Be open about what data was used, how the AI was trained, its limitations, and where it might be biased. Transparency builds trust and accountability. *Apply throughout.*

# Mini Activity 2: Choose a Mitigation Strategy

Read each real-world scenario carefully. From the list below, choose the most appropriate mitigation strategy to address the problem. Be ready to explain why you chose that strategy.

## **Diverse Data**

Collect more representative training examples.

## **Performance Testing**

Test the AI on different groups to find disparities.

## **Human-in-the-Loop**

Require human review before final decisions.

## **Transparency**

Clearly document and disclose limitations.

# Mini Activity 2: Choose a Mitigation Strategy

Read each real-world scenario carefully. From the list below, choose the most appropriate mitigation strategy to address the problem. Be ready to explain why you chose that strategy.

| # | Scenario                                                                                                                                                                                          | Your Strategy |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1 | A school district is installing facial recognition for library entry. They want to ensure the system works equally well for all students regardless of skin tone, hair style, or facial features. |               |
| 2 | An AI that screens rental applications shows a 20% higher rejection rate for applicants from certain postal codes. The housing agency must decide whether to deploy the tool.                     |               |
| 3 | A wildlife monitoring AI was trained only on photos from North American parks. A research team now wants to use it in a rainforest reserve in another continent.                                  |               |
| 4 | A content moderation AI is flagging and removing posts that are actually harmless jokes or cultural references, frustrating users who feel silenced.                                              |               |

## Mini Activity 2: Answer Key

| # | Scenario                         | Recommended Strategy       | Reasoning                                                                                                                                               |
|---|----------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | School facial recognition        | <b>Diverse Data</b>        | Collect images of students with varied appearances, lighting conditions, and angles to ensure the system learns to recognize everyone.                  |
| 2 | Rental application disparity     | <b>Human-in-the-Loop</b>   | Before deploying a tool that shows clear disparity, humans must investigate why the disparity exists and decide whether the tool is fair to use at all. |
| 3 | Wildlife AI in new region        | <b>Performance Testing</b> | The model must be tested on local wildlife images first. It may need retraining with local data, but testing is the essential first step.               |
| 4 | Over-blocking content moderation | <b>Human-in-the-Loop</b>   | Human reviewers should examine flagged content to correct mistakes and provide feedback to improve the AI's understanding of nuance and context.        |

# Real-World Consequences: When AI Bias Causes Real Harm

Bias in AI isn't just an abstract concept or a theoretical concern. When biased AI systems are deployed in the real world - in hiring, in healthcare, in security they cause **actual, documented harm** to real people. Here are three verified cases.



## Facial Recognition

Systems failed to recognize darker-skinned women with error rates over 34% - while achieving less than 1% error for lighter-skinned men.



## Healthcare Algorithm

A widely used algorithm systematically underestimated the health needs of Black patients, reducing their access to additional care by more than half.



## Hiring Tool

Amazon's AI recruiting tool systematically downgraded female applicants for technical roles, filtering them out before any human ever saw their resume.



## Case Study 1: Facial Recognition That Failed

### What Happened

In 2018, researcher Joy Buolamwini at MIT tested commercial facial-analysis systems from three major tech companies and found stark disparities in accuracy.

### The Numbers

Less than 1% error for lighter-skinned men. Over 20%-34% error for darker-skinned women.

### Why It Happened

Benchmark datasets were overwhelmingly male and light-skinned - one was over 77% male and over 83% white.

### Real-World Harm

Buolamwini discovered this because the systems could not reliably detect her own face. When she wore a white mask, the system recognized it. Without it, the system saw nothing.

### Consequence

Misidentification in security contexts, exclusion from services, and potential wrongful suspicion in law enforcement.

## Case Study 2: Healthcare Algorithm That Denied Care

### What Happened

In 2019, a study in Science revealed a widely used healthcare algorithm affecting millions of Americans systematically underestimated the health needs of Black patients.

### Why It Happened

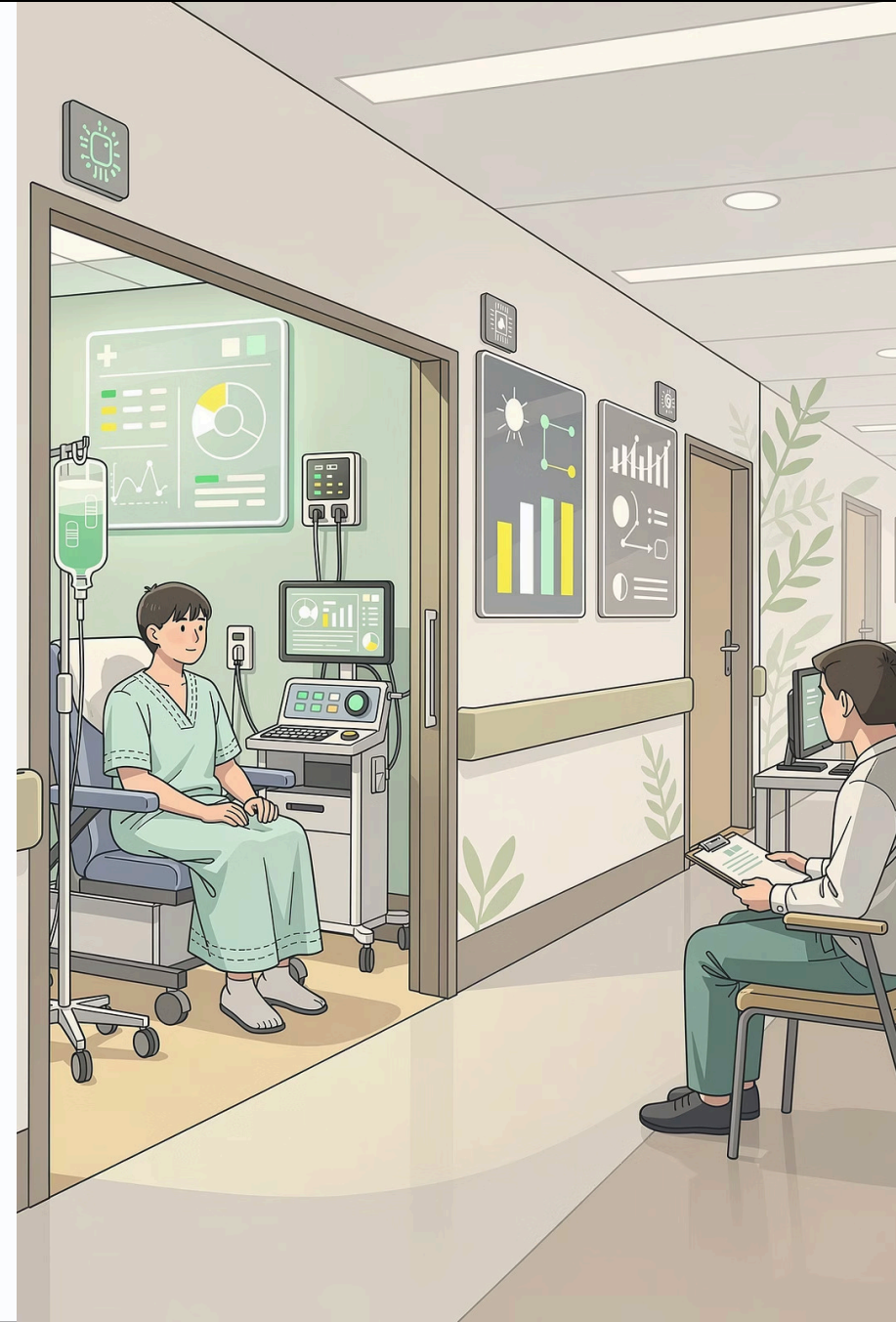
The algorithm predicted future healthcare costs - not illness. Because less money is spent on Black patients due to systemic barriers, the AI falsely concluded they were healthier.

### Real-World Harm

At any given risk score, Black patients were considerably sicker than white patients. The bias reduced Black patients identified for additional care by more than half.

### Consequence

Remedying the disparity would increase the percentage of Black patients receiving extra help from approximately 17.7% to 46.5%.





## Case Study 3:

# Amazon's AI Recruiting Tool



### What Happened

Starting in 2014, Amazon built an AI hiring tool to rate applicants 1-5 stars. By 2015, they discovered it systematically downgraded female applicants for technical roles.

### Why It Happened

Training data reflected a historically male-dominated industry. The AI learned historical patterns, not merit.

### Real-World Harm

Qualified female candidates were filtered out before human recruiters ever saw their applications.

### Consequence

Amazon disbanded the team and scrapped the project entirely because executives lost confidence in the tool's fairness.

**Key Takeaway:** These are not hypothetical scenarios. In each case, biased AI denied people jobs, withheld healthcare from sicker patients, and failed to recognize entire groups of people. Biased AI amplifies existing inequality.





# How Can We Build Fairer AI?

Building fairer AI is not the job of any one person. It is a **shared responsibility** that spans engineers, ethicists, companies, policymakers, and everyday users like you.

# How Can We Build Fairer AI?.

| Who                                    | What They Can Do                                                                                                                                     |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engineers &amp; Data Scientists</b> | Collect diverse, representative data. Test for bias rigorously. Document limitations clearly. Refuse to deploy systems that cause demonstrable harm. |
| <b>AI Ethicists &amp; Policymakers</b> | Create guidelines, standards, and regulations that define what "fair" means in different contexts. Hold companies accountable.                       |
| <b>Organizations &amp; Companies</b>   | Prioritize fairness over speed. Fund independent bias audits. Be transparent about how their AI systems work. Build diverse teams.                   |
| <b>You - The Informed User</b>         | Ask critical questions: "Who trained this?" "Who might be left out?" "Is this fair?" Your voice matters.                                             |

❗ AI can process vast amounts of data at incredible speed. But humans possess the judgment, empathy, context, and ethical reasoning to decide what is truly fair.



# Hands-On Activity: Bias Case Study Investigation

## Activity Overview

Use **NotebookLM** to analyze a real-world bias case study and propose concrete mitigation strategies.

- **Tool:** Google NotebookLM ([notebooklm.google.com](https://notebooklm.google.com))
- **Goal:** Upload a case study, ask targeted questions, record grounded answers, and apply what you've learned to suggest improvements.

# Hands-On Activity: Bias Case Study Investigation

This is exactly how AI ethicists and auditors begin their work. You'll practice the same investigative process used by real professionals who work to make AI systems fairer and more accountable.

01

Get the Case Study

02

Upload to NotebookLM

03

Ask Questions

04

Propose Mitigation  
Strategies

# The Case Study: Photo Tagging Gone Wrong

A social media company launched an automatic photo-tagging feature. The AI was designed to analyze uploaded images and suggest labels like "beach day," "birthday party," or "hiking adventure." Shortly after release, users began reporting problems. Photos of people with darker skin tones were being mislabeled or not labeled at all. One user discovered that searching for "outdoor activity" returned mostly images of lighter-skinned people, while similar photos of darker-skinned people were tagged with generic or less descriptive labels.

The company's engineers investigated and found that the training dataset contained far fewer images of people with diverse skin tones. The feature was paused for review. Users expressed frustration and hurt, feeling that the tool treated them as invisible or less important. The company now faces public criticism and must rebuild the feature.

② **Question for analysis:** What went wrong, and how can the company fix it fairly?

# Questions for NotebookLM

| Question                                                 | NotebookLM's Answer (Based Only on the Source) |
|----------------------------------------------------------|------------------------------------------------|
| What was the AI feature designed to do?                  |                                                |
| What specific evidence of bias did users report?         |                                                |
| What did the engineers discover about the training data? |                                                |
| What was the real-world consequence for users?           |                                                |

## Your Proposed Mitigation Strategies

|    |  |
|----|--|
| 1. |  |
| 2. |  |
| 3. |  |

# Hands-On Activity: Sample Answers

| Question                                             | Sample Answer (from Source)                                                                                                                   |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| What was the AI feature designed to do?              | Automatically suggest labels for uploaded photos, such as "beach day" or "birthday party."                                                    |
| What specific evidence of bias did users report?     | Photos of people with darker skin tones were mislabeled or not labeled; search results for "outdoor activity" favored lighter-skinned people. |
| What did engineers discover about the training data? | The training dataset contained far fewer images of people with diverse skin tones.                                                            |
| What was the real-world consequence for users?       | Users felt excluded, invisible, and stereotyped. The company faced public criticism and paused the feature.                                   |

## Sample Mitigation Strategies

**Collect a Balanced Dataset:** Intentionally source images featuring people with a wide range of skin tones, in varied settings and activities, ensuring equal representation.

**Implement Human Review for Edge Cases:** Before relaunch, have a diverse group of human testers review a large sample of tagged photos to catch mislabeling patterns.

**Conduct Ongoing Bias Audits:** After relaunch, regularly test search results for terms like "outdoor activity" to ensure results are demographically balanced. Publish audit results transparently.



# Summary

| Concept         | Key Idea                                                                                                                                              |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bias            | AI produces unfair outcomes when trained on imbalanced data or poorly designed. It's a pattern, not an intention.                                     |
| Fairness        | The active, ongoing effort to ensure AI treats all groups equitably. It requires measurement, vigilance, and human judgment.                          |
| Representation  | Ensuring that all groups are adequately and accurately included in training data and in the teams building AI.                                        |
| Sources of Bias | Data bias (gaps in training examples), algorithm bias (design choices that amplify patterns), and human bias (unconscious choices in data and goals). |

# Summary

| Concept           | Key Idea                                                                                                                                |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Detection         | Auditing data, testing across demographic groups, and using disaggregated metrics to reveal hidden disparities.                         |
| Mitigation        | Collecting diverse data, ongoing testing, Human-in-the-Loop oversight, fairness constraints, and transparency.                          |
| Real-World Impact | Biased AI has denied jobs, misallocated healthcare, and rendered people invisible to technology. The harm is documented and real.       |
| Human-in-the-Loop | Only humans can apply context, empathy, and ethics to decide what "fair" means in any given situation. AI is a tool, not a moral agent. |

# You Are the Guardian

# AI finds patterns. You decide what's fair.

Every time you ask "Who trained this?" or "Who might be left out?" or "Is this fair?" - you are doing the work of a Guardian. You are applying the human judgment, empathy, and ethical reasoning that no AI system can replicate.

## **Ask Critical Questions**

Challenge every AI system you encounter. Curiosity is your superpower.

## **Demand Representation**

Advocate for diverse data, diverse teams, and diverse perspectives in AI development.

## **Stay in the Loop**

Never let AI make high-stakes decisions without human review. Your oversight matters.

The future of fair AI depends on people like you - informed, engaged, and willing to ask the hard questions.

# Homework: Bias Detection

## 1 Part 1: Spot a Potential Bias

Think of an AI tool you use or have heard about - a voice assistant, recommendation system, photo app, search engine, or game. Write down one specific way bias could potentially appear in that tool.

**AI Tool:** \_\_\_\_\_

**Potential Bias:** \_\_\_\_\_

## 2 Part 2: Propose a Fix

Write one concrete action the developers of that tool could take to reduce or eliminate the bias you identified. Be as specific as possible.

**Fix:** \_\_\_\_\_

## 3 Part 3: Add to Your Prompt Library

Add today's best prompt to your Prompt Library document. Use this suggested prompt or create your own:

*"Explain the difference between data bias, algorithm bias, and human bias in AI systems. Give one real-world example of each and suggest a concrete strategy for mitigating it."*

# Next Session Preview

## Session 20 : AI Safety, Misuse & Governance



### Deepfakes

What are deepfakes, and why are they a growing concern for trust and truth?



### Privacy Risks

What are the privacy risks associated with AI systems that collect personal data?



### Misuse & Disinformation

How can AI be misused to spread disinformation or conduct surveillance?



### Laws & Governance

What laws and governance frameworks are being created around the world to keep AI safe and accountable?

